

Passivity-Based Control with Physics-Informed Structured Learning for Robot Manipulators

Master's Thesis Proposal

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1 Motivation and Problem Statement

Robot manipulators are nonlinear dynamical systems subject to parametric uncertainty, unmodeled dynamics, friction, external disturbances and payload variations. Their Euler–Lagrange structure makes it possible to construct storage functions associated with the mechanical energy and to design controllers whose stability is established through passivity arguments. The performance of these schemes, however, depends critically on the quality of the nominal model: in the presence of significant modeling errors, feedforward compensation is insufficient.

There are two classical answers to this problem, both with well-known limitations. Adaptive control [22] guarantees convergence of the tracking error, but requires the uncertainty to be *linear in the parameters* with respect to a known regressor; effects such as Stribeck friction, transmission flexibility or *backlash* do not admit such a parameterization. Sliding mode control guarantees robustness against arbitrary bounded uncertainty, but the gains must be sized for the *worst case* of the total uncertainty, which induces chattering, high control effort and amplification of measurement noise.

Neural compensation of PD and PID loops occupies the intermediate position and has a mature Lyapunov theory behind it. In that line of work the network is a *feedforward* block added to an existing loop: the closed loop is proved uniformly ultimately bounded with a residual radius proportional to the network approximation error rather than to the whole unmodeled torque, the learning law is derived from the Lyapunov derivative rather than from backpropagation, and only the output layer is adapted so that the parameterization is linear and free of local minima. The limitation of that family is equally clear: the error converges to a *ball*, never to zero, and shrinking the ball requires large derivative gains. The lesson to be transferred is that the residual set can be *shrunk* by learning but must be *removed* by an additional mechanism with integral character.

The present proposal starts from the observation that these limitations are complementary. If a *structured* learning module captures the systematic and repeatable part of the uncertainty, the robust controller no longer has to face the total uncertainty but only a residual of considerably smaller magnitude, which allows the sliding mode gains to be reduced; and the second-order sliding mode algorithm, being a nonlinear integrator, removes the residual set that learning alone can only shrink. The central research problem is therefore twofold:

- (P1) **Structural compatibility.** An arbitrary learned compensation modifies the energy balance of the closed loop and may destroy the passivity properties used to prove stability. Under which restrictions on the network parameterization — and on its online adaptation — is the passive structure of the closed-loop system preserved?

(P2) **Quantification of the benefit.** Is it possible to establish an explicit relation between the quality of the neural approximation and the minimum admissible gains of the robust algorithm, so that the contribution of learning is quantifiable and not merely empirical?

1.1 Three levels of model knowledge

Table 1 places the proposal with respect to the two classical answers and to the neural PD/PID literature. The architecture proposed here is the fourth column: the nominal passivity-based loop is kept intact, a structured learning module absorbs the systematic part of the residual, and a second-order sliding mode action removes what remains.

	Computed torque	PBC + SMC	PBC + learning	This proposal
model required	exact M, C, g, F	nominal	nominal	nominal
handles unstructured	no	yes	no	yes
gains sized for	—	total Δ	—	residual $\tilde{\Delta}$
error converges to	0	0 (finite time)	ball $\propto \bar{\epsilon}$	0 (finite time)
control signal	continuous	chattering	continuous	continuous

Table 1: Levels of model knowledge and what each architecture can guarantee.

1.2 Formulation

Consider a rigid manipulator with n degrees of freedom,

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) + F(\dot{q}) + \tau_d(t) = \tau, \quad (1)$$

where $q \in \mathbb{R}^n$ are the joint coordinates, $M(q)$ is the inertia matrix, $C(q, \dot{q})\dot{q}$ collects the Coriolis and centrifugal terms, $g(q)$ is the gravitational torque, $F(\dot{q})$ is friction and $\tau_d(t)$ represents external disturbances. The model satisfies the structural properties

$$\lambda_m I \preceq M(q) \preceq \lambda_M I, \quad 0 < \lambda_m \leq \lambda_M < \infty, \quad (2)$$

$$x^\top (\dot{M}(q) - 2C(q, \dot{q}))x = 0, \quad \forall x \in \mathbb{R}^n, \quad \dot{M} = C + C^\top, \quad (3)$$

$$\dot{q}^\top F(\dot{q}) \geq 0, \quad (4)$$

$$\|C(q, \dot{q})x\| \leq k_c \|\dot{q}\| \|x\|, \quad \|g(q)\| \leq k_g, \quad \|F(\dot{q})\| \leq k_f(1 + \|\dot{q}\|). \quad (5)$$

Property (3) is the one that makes the map $\tau \mapsto \dot{q}$ passive with a storage function equal to the total energy, and it is also the one that must be preserved when the learning module is introduced. The Christoffel form $\dot{M} = C + C^\top$ is the version of (3) that will be imposed on the learned inertia. The growth bounds (5) are not decorative: they are what converts a bound on the network output into a bound on $\frac{d}{dt}\tilde{\Delta}$, and therefore what closes the gain condition of the robust action.

Let $q_d(t)$ be a bounded desired trajectory of class C^2 . Define

$$e = q - q_d, \quad \dot{q}_r = \dot{q}_d - \Lambda e, \quad s = \dot{q} - \dot{q}_r = \dot{e} + \Lambda e, \quad \Lambda = \Lambda^\top \succ 0. \quad (6)$$

The variable s has relative degree one with respect to τ , which is essential: it allows a second-order algorithm producing a *continuous* control signal to be applied.

Remark 1 (Origin of the filtered error). *The signal s is not an arbitrary choice. In the Lyapunov analysis of PD regulation with neural compensation, the natural energy function yields a derivative*

that bounds $\dot{q}$ but says nothing about e , and the adaptation stops precisely when the arm settles — exactly where the steady-state bias lives. Adding the cross term $\alpha e^\top M(q)\dot{q}$ to the storage function injects $-\alpha e^\top K_p e$ into $\dot{V}$ and makes the combination $\dot{q} + \alpha e$ appear in the learning law. The same object, with a general matrix Λ in place of αI , is (6). It is therefore the signal that simultaneously drives the control, the adaptation and the sliding surface.

With a nominal model $(\hat{M}, \hat{C}, \hat{g})$, the proposed controller has the structure

$$\tau = \underbrace{\hat{M}(q)\ddot{q}_r + \hat{C}(q, \dot{q})\dot{q}_r + \hat{g}(q) - K_D s}_{\tau_{\text{pbc}}} + \underbrace{\hat{\Delta}_\theta(q, \dot{q}, \dot{q}_r, \ddot{q}_r)}_{\tau_\ell} + \underbrace{\tau_{\text{sta}}}_{\text{robust action}}, \quad (7)$$

where $K_D = K_D^\top \succ 0$. Substituting (7) into (1) yields the error dynamics

$$M(q)\dot{s} = -C(q, \dot{q})s - K_D s - \tilde{\Delta} + \tau_{\text{sta}}, \quad \tilde{\Delta} := \Delta - \hat{\Delta}_\theta, \quad (8)$$

with the total uncertainty

$$\Delta = \tilde{M}(q)\ddot{q}_r + \tilde{C}(q, \dot{q})\dot{q}_r + \tilde{g}(q) + F(\dot{q}) + \tau_d(t), \quad (\tilde{\cdot}) = (\cdot) - (\hat{\cdot}). \quad (9)$$

Remark 2. The choice of expressing Δ in terms of $\dot{q}_r$ and $\ddot{q}_r$ — and not of $\dot{q}$ and $\ddot{q}$ — is deliberate: Δ does not depend on the measured acceleration, but only on known signals and on the state. This is indispensable both for the implementation and for the generation of training data.

1.3 The baseline to beat

Without the learning module and without the robust action, (8) reduces to $M\dot{s} = -Cs - K_D s - \Delta$. Bounding the cross term with Young's inequality on $V = \frac{1}{2}s^\top M(q)s$ gives $\dot{V} \leq -s^\top (K_D - \Lambda_1)s + \bar{\eta}_1$ with $\bar{\eta}_1 = \sup_t \Delta^\top \Lambda_1^{-1} \Delta$, hence

$$\limsup_{t \rightarrow \infty} \|s(t)\| \leq \sqrt{\bar{\eta}_1 / \lambda_{\min}(K_D - \Lambda_1)} =: \rho_1. \quad (10)$$

The radius ρ_1 is set by the *entire* unmodeled torque and can only be reduced by raising K_D , at the price of noise amplification and actuator saturation. Learning replaces Δ by $\tilde{\Delta}$ in $\bar{\eta}_1$ and therefore shrinks ρ_1 at the same gain; the robust action then drives the residual set to zero in finite time. These are the two distinct mechanisms the proposal aims to separate and quantify.

1.4 Structured Decomposition of the Residual

A conceptual contribution of the proposal is to recognize that Δ is *not* entirely of Euler–Lagrange type, and that only a part of it admits learning with passivity guarantees. The following decomposition is proposed:

$$\Delta = \underbrace{\Delta_{\text{el}}}_{\text{EL structure}} + \underbrace{\Delta_{\text{dis}}}_{\text{dissipative}} + \underbrace{\Delta_{\text{ext}}}_{\text{unstructured}}, \quad (11)$$

where $\Delta_{\text{el}} = \tilde{M}\ddot{q}_r + \tilde{C}\dot{q}_r + \tilde{g}$ retains the Lagrangian structure, $\Delta_{\text{dis}} = F(\dot{q})$ satisfies (4) and therefore *contributes* to dissipation, and $\Delta_{\text{ext}} = \tau_d(t)$ has no exploitable structure. The learning module targets the first two terms; the third is necessarily left to the robust action. This separation delimits precisely what learning can and cannot contribute, and avoids the claim — frequent in the literature and untenable — that a neural network can replace robustness.

2 Relation to Prior Work and Delimitation of the Contribution

Both components of the proposed architecture have well-established prior art. Stating this explicitly, and early, is a requirement of intellectual honesty and also the only way to make the actual contribution visible. Table 2 lists the labels that are *already taken*.

Component / label	Prior art that owns it
Sliding surface $s = \dot{e} + \Lambda e$ on an Euler–Lagrange system	[23, 22] — this <i>is</i> the classical robot sliding surface.
Passivity-based control combined with sliding modes	Integral sliding manifolds for systems with unmatched perturbations [2], and the PBC + ISM line that follows it.
Structure-preserving (“physics-informed”) networks for mechanical systems	Deep Lagrangian Networks [11]; Lagrangian and Hamiltonian neural networks [3, 5]. The constructions $\hat{M} = LL^\top$, Christoffel $\hat{C}$ and $\nabla_q \hat{U}$ are literally DeLaN.
Spectrally normalized network whose Lipschitz constant enters the stability proof	Neural Lander [21].
Lyapunov-derived learning law, no backpropagation, no local minima	[20, 10, 27].
Neural compensation added to an existing industrial PD/PID loop	[29, 30].

Table 2: Components of the architecture and the prior work that already owns each label.

Consequently, “passivity-based sliding mode control” and “physics-informed neural network” are *not*, by themselves, contributions of this proposal: each is a named body of literature, and presenting their juxtaposition as novel would reduce the answer to “what is new?” to “the combination”, which is the weakest possible answer. What is new is the *interface* between the two, and it consists of the following three items.

(N1) A derivative certificate along closed-loop trajectories

Existing Lipschitz-certified learning-based control [21] bounds the Lipschitz constant of the learned *map* and uses it in an ISS-type argument. The super-twisting algorithm requires something strictly harder: a bound on $\|\frac{d}{dt}\tilde{\Delta}\|$ *along the closed-loop trajectories*, where

$$\frac{d}{dt}\tilde{\Delta} = \underbrace{\frac{d\Delta}{dt}}_{\text{plant, via (5)}} - \underbrace{\frac{\partial(\hat{W}\Sigma_\theta)}{\partial x}}_{\text{network}} \dot{x} - \underbrace{\dot{\hat{W}}\Sigma_\theta}_{\text{adaptation}}. \quad (12)$$

The third term does not exist in any offline-learning formulation, and it is bounded by the learning law itself: $\|\dot{\hat{W}}\| \leq \lambda_{\max}(K_w)\|s\|\|\Sigma_\theta\|$ from (17), with $\|\Sigma_\theta\|$ bounded by projection. Closing (12) is what makes Assumption 1 a theorem rather than a hypothesis, and it is the sharpest single novelty of the proposal.

(N2) Structure-preserving *adaptation*, not structure-preserving *approximation*

Deep Lagrangian networks [11] provide structure for a network that is *fitted offline and then frozen*; the structural properties are properties of a fixed function. The claim here is different: the admissible set $\mathcal{W}$ is constructed so that the structural properties hold for *every* $\hat{W} \in \mathcal{W}$, and the projected law (17) keeps $\hat{W}(t) \in \mathcal{W}$ for all t . Passivity therefore holds along every trajectory *while the weights are still moving*, which is what an online law requires and what a frozen-network result does not provide. Structure-preserving adaptation is the new object.

(N3) A negative result, and a positive one that is quantified

Much of the neural sliding-mode literature reports improved tracking accuracy from learning while using an ideal sliding mode in the proof; the two statements are incompatible, as Remark 6 shows. Stating the limitation correctly — learning cannot improve ideal asymptotic accuracy — and then locating the benefit where it is provable, namely in the map $(\delta_\theta, L_\theta) \mapsto (k_1^{\min}, k_2^{\min})$ and in the sampled accuracy $|s| = O(k^2 h^2)$, is itself a contribution in a field that currently has the claim backwards.

Remark 3 (The claim in one sentence). *The contribution is not the architecture but the certified interface between learning and robustness: a structure-preserving parameterization whose admissible set makes online adaptation passivity-compatible, and whose spectral bounds yield an a priori bound on $\|\frac{d}{dt}\tilde{\Delta}\|$ — the quantity the super-twisting gain condition requires and the neural sliding-mode literature assumes.*

3 Hypothesis

If the learning module is parameterized so as to reproduce by construction the structural properties (2)–(4) of the system, and if its online adaptation is derived from the Lyapunov derivative so that the corresponding trace term cancels exactly, then (i) the closed loop retains the passivity properties of the nominal controller for every admissible value of the weights, and (ii) the Lipschitz bound of the uncompensated residual admits an explicit certificate that translates into a quantifiable reduction of the minimum admissible gains of the robust algorithm, without loss of the finite-time convergence guarantees.

4 Objectives

General objective

To develop and analyze a control architecture for robot manipulators that integrates passivity-based control, physics-informed structured learning and a robust second-order sliding mode action, establishing formal conditions for stability, for preservation of the energy properties and for finite-time convergence of the compensation residual, as well as a quantitative characterization of the benefit contributed by learning.

Specific objectives

1. Design the nominal passivity-based controller (7) and develop its stability analysis, characterizing the passive map $\tilde{\Delta} \mapsto s$, its dissipation index and the baseline residual radius (10).
2. Formulate a learning module whose parameterization imposes *by construction* — and not through penalties in the cost function — the symmetry and positive definiteness of the estimated inertia, the skew-symmetry of $\dot{\hat{M}}_\theta - 2\hat{C}_\theta$, the gradient character of the gravitational term and the monotonicity of the friction term.
3. Formulate the offline training problem in an *acceleration-free* form, based on the filtered or integral version of (1), and define the persistent excitation protocol for data collection.
4. Derive an *online* adaptation law for the output layer of the module from the Lyapunov derivative of the closed loop, such that the associated trace term cancels exactly and the structural

constraints are preserved for every weight in the admissible set; prove that the resulting closed loop is output strictly passive, and guarantee uniform boundedness of $\hat{\Delta}_\theta$ by projection, so that a poorly trained module degrades performance without compromising stability.

5. Establish the actuator-compatibility condition relating the certified saturation level $\bar{\Delta}$ of the module to the available torque, so that the robust action retains authority when the module is at its bound.
6. Design the Super-Twisting robust action on the residual $\tilde{\Delta}$, explicitly addressing the coupling induced by $M^{-1}(q)$ either through a multivariable formulation or through decoupling bounds derived from (2).
7. Establish a Lipschitz bound $\|\frac{d}{dt}\tilde{\Delta}\| \leq L_\theta$ certified from the network parameterization (spectral normalization and projection) together with the growth bounds (5), and derive the explicit relation $(\delta_\theta, L_\theta) \mapsto (k_1^{\min}, k_2^{\min})$ between approximation quality and minimum admissible gains.
8. Characterize the benefit of learning in the metrics where it is theoretically sustainable: magnitude of the gains, control effort and total variation of the control, duration of the reaching phase and real sliding accuracy under sampling.
9. Experimentally validate the strategy on a robotic platform, comparing the nominal controller, the controller with learned compensation, the nominal controller with full-gain STA and the complete architecture.

5 Methodology

Every element of the architecture is taken from an identified piece of prior work and then modified for a specific purpose. Table 3 states the correspondence; each subsection below repeats it as a short *Provenance* paragraph, so that the reader can separate at a glance what is imported from what is added.

5.1 Structured Learning Module

Provenance. The four constructions are those of deep Lagrangian networks [11], with the Hamiltonian and Lagrangian network variants [5, 3] as antecedents. *What is modified.* Those works learn a *simulator*; here the module learns the control residual $\Delta_{\text{el}} + \Delta_{\text{dis}}$ of (11), is expressed in terms of $\dot{q}_r, \ddot{q}_r$ so that it never requires a measured acceleration, and is saturated at a certified $\bar{\Delta}$. Only then can the structural properties be used as hypotheses of a Lyapunov argument rather than as qualitative model priors.

Unlike a *physics-informed* network in the sense of [18], where physical knowledge is introduced as a penalty on the residual of a PDE in the cost function, here the structure is imposed in the *architecture* itself, following the line of deep Lagrangian networks [11]. The parameterization is

$$\hat{\Delta}_\theta = \hat{M}_\theta(q) \ddot{q}_r + \hat{C}_\theta(q, \dot{q}) \dot{q}_r + \nabla_q \hat{U}_\theta(q) + \hat{D}_\theta(\dot{q}) \dot{q}, \quad (13)$$

with the following constructions:

- $\hat{M}_\theta(q) = L_\theta(q) L_\theta^\top(q) + \epsilon I$, with L_θ lower triangular with positive diagonal. This automatically guarantees $\hat{M}_\theta = \hat{M}_\theta^\top \succ 0$ and the analogue of (2).

Element	Comes from	How it is modified here
Filtered error $s = \dot{e} + \Lambda e$; nominal PBC law	[23, 22, 15]	s is made to serve three roles at once — control, adaptation and sliding variable — so the passive storage and the STA Lyapunov function share a variable
Structured module $\hat{M} = LL^\top$, Christoffel $\hat{C}$, $\nabla_q \hat{U}$, $\hat{D}$	[11, 3, 5]	the target is the control residual $\Delta_{\text{el}} + \Delta_{\text{dis}}$, not the full dynamics; output saturated at a certified $\bar{\Delta}$ so the structure is usable inside a stability proof
Acceleration-free (filtered / integral) regression	[1, 12]	the same operator $\mathcal{F}$ is reused at run time as the velocity estimator, so training and deployment see identical signals
Lyapunov learning law with exact trace cancellation	[20, 10, 27, 29]	the law acts on a <i>typed</i> regressor and is projected onto $\mathcal{W}$, so the structural properties survive the adaptation itself (N2)
e -, σ -modification, dead zone, projection	[6]	projection is promoted from safeguard to <i>certificate ingredient</i> : it supplies the bound on $\ \dot{W}\ _F$ used to compute L_θ
Anti-windup condition on the learned torque	[29]	restated as a three-way actuator budget among τ_{pbc} , $\bar{\Delta}$ and the authority ν_{max} reserved for the robust action
Super-twisting algorithm and strict Lyapunov function	[8, 9, 13]	driven by a residual whose Lipschitz bound is <i>certified</i> rather than assumed; storage function composed with the passive and parameter terms
Multivariable STA under a matrix input gain	[14]	the coupling $M^{-1}(q)$ is bounded through (2), with $\kappa = \lambda_M/\lambda_m$ appearing explicitly in the gain constants
Spectral normalization inside a stability certificate	[21]	the certified object is the <i>time derivative</i> along the closed loop, including the adaptation term $\dot{W}\Sigma_\theta$ absent from offline formulations (N1)
Variable-gain STA	[4]	used as the declared fallback when the architectural certificate is too conservative

Table 3: Methodological roadmap: element, source, and the modification introduced here.

- $\hat{C}_\theta$ is obtained from $\hat{M}_\theta$ through the Christoffel symbols, which makes $\dot{\hat{M}}_\theta - 2\hat{C}_\theta$ skew-symmetric *by construction*, preserving (3).
- The gravitational term is obtained as the gradient of a learned scalar potential $\hat{U}_\theta(q)$, which guarantees its conservative character.
- $\hat{D}_\theta(\dot{q}) \succeq 0$ is parameterized with nonnegative weights and monotone activations, so that $\dot{q}^\top \hat{D}_\theta(\dot{q}) \dot{q} \geq 0$, respecting (4).
- The output of (13) is saturated at a bound $\bar{\Delta}$ obtained from the physical specifications of the manipulator.

5.2 Acceleration-Free Offline Training

Provenance. Filtering or integrating the model to eliminate $\ddot{q}$ is standard in robot parameter identification [1, 12]; the Fourier-series excitation design is likewise classical. *What is modified.* The same

operator $\mathcal{F}$ is reused at implementation time as the velocity estimator of (20), so the module is deployed on exactly the signals it was fitted on; and the regularizer carries the spectral normalization that later becomes part of the certificate, rather than being a generic weight penalty.

The residual Δ is not directly measurable, since it would involve $\ddot{q}$. Applying the stable first-order filter $\lambda/(p + \lambda)$ to (1), or else integrating over a window $[t - T, t]$ and integrating the inertial term by parts by means of (3), one obtains an equivalent expression involving only the measured q , $\dot{q}$ and τ . The training cost takes the form

$$J(\theta) = \sum_k \|\mathcal{F}\{\Delta\}_k - \mathcal{F}\{\hat{\Delta}_\theta\}_k\|^2 + \gamma \mathcal{R}(\theta), \quad (14)$$

where $\mathcal{F}\{\cdot\}$ denotes the filtering operator and $\mathcal{R}$ a regularizer that includes the spectral normalization of each layer. Data will be collected with persistent excitation trajectories designed by means of finite Fourier series, with independent training, validation and test partitions.

Remark 4 (What persistent excitation is and is not needed for). *Persistent excitation is not required for closed-loop stability: the Lyapunov argument of Section 5.3 only needs the trace term to cancel. It is required for parameter convergence, that is, for the estimate δ_θ of the residual magnitude to be meaningful on the whole operating set rather than on the visited trajectories. Since the quantified benefit of the proposal rests on δ_θ and L_θ , PE is a requirement of the certificate, not of the stability proof — a distinction that is often blurred in the neural control literature.*

5.3 Online Adaptation and Exact Trace Cancellation

Provenance. The gradient-like law with exact trace cancellation is the standard device of Lyapunov-based neural control [20, 10, 27, 29]. *What is modified.* In those works the regressor $\sigma(x)$ is an unstructured basis, so the compensation has no physical type and the structural properties play no role. Here the regressor is *typed* — inertial, Coriolis, potential, dissipative — and the adapted weights are confined to $\mathcal{W}$, so the Euler–Lagrange properties hold for every admissible weight and therefore along every trajectory while the weights are still moving. This is contribution (N2).

Freezing the structure-generating layers after offline training leaves a regressor $\Sigma_\theta(x) \in \mathbb{R}^{n \times m}$ formed by the four channels of (13), with $x = [q^\top, \dot{q}^\top, \ddot{q}_r^\top, \ddot{q}_r^\top]^\top$. Writing the module as

$$\hat{\Delta} = \hat{W} \Sigma_\theta(x), \quad \Delta_{\text{el}} + \Delta_{\text{dis}} = W^* \Sigma_\theta(x) + \varepsilon(x), \quad \|\varepsilon(x)\| \leq \bar{\varepsilon}, \quad (15)$$

the parameterization is linear in the adapted weights, so a Lyapunov learning law exists with no back-propagation and no local minima. Restricting $\hat{W}$ to the admissible set $\mathcal{W} = \{\hat{W} \text{ block diagonal, } \hat{W} \succeq 0, \|\hat{W}\|_F \leq W_{\max}\}$ keeps each channel's structural property valid for every admissible weight: a positive scaling of $\hat{M}_\theta$ leaves it positive definite and rescales $\hat{C}_\theta$ consistently, a positive scaling of $\nabla_q \hat{U}_\theta$ is the gradient of a scaled potential, and a positive scaling of $\hat{D}_\theta$ remains positive semidefinite.

Taking the composite storage function $V_\ell = \frac{1}{2} s^\top M(q) s + \frac{1}{2} \text{tr}(\tilde{W}^\top K_w^{-1} \tilde{W})$ with $\tilde{W} = \hat{W} - W^*$, using (3) and the identity $s^\top \tilde{W} \Sigma = \text{tr}(\tilde{W}^\top s \Sigma^\top)$, the derivative becomes

$$\dot{V}_\ell = -s^\top K_D s - s^\top \varepsilon + s^\top \tau_{\text{sta}} + \text{tr}[\tilde{W}^\top (K_w^{-1} \dot{\hat{W}} + s \Sigma^\top(x))]. \quad (16)$$

The trace term vanishes identically for the gradient-like law

$$\boxed{\dot{\hat{W}} = -K_w s \Sigma^\top(x)}, \quad K_w = K_w^\top \succ 0, \quad (17)$$

projected onto $\mathcal{W}$. The learned block therefore cannot inject energy into the loop: the passivity of the nominal design is preserved for every trajectory, not merely at the converged weights. This is the formal answer to problem (P1).

5.4 Robust Modifications and Certified Bounds

Provenance. The e - and σ -modifications, the dead zone and parameter projection are the classical robust-adaptation toolbox [6], used in neural control to prevent drift. *What is modified.* Projection is not treated as a safeguard here: the bound $\|\hat{W}\|_F \leq W_{\max}$ it enforces is one of the three terms of the derivative certificate (12), so removing it would not merely risk drift — it would invalidate the gain condition.

Pure integral adaptation drifts under measurement noise and unmodeled excitation. The standard remedies will be evaluated, each with its own contribution to the certificate:

- e -modification, $\dot{\hat{W}} = -K_w(s\Sigma^\top + \kappa\|s\|\hat{W})$, and σ -modification, $\dot{\hat{W}} = -K_w(s\Sigma^\top + \kappa\hat{W})$: each contributes $-\kappa \operatorname{tr}(\tilde{W}^\top \hat{W}) \leq -\frac{\kappa}{2}\|\tilde{W}\|_F^2 + \frac{\kappa}{2}\|W^*\|_F^2$ to $\dot{V}_\ell$, giving joint boundedness of $(s, \tilde{W})$.
- Dead zone: adapt only when $\|s\|$ exceeds the noise floor.
- Projection onto $\mathcal{W}$: bounds $\hat{\Delta}$ uniformly *and* bounds the Lipschitz constant of the module, which is what the robust action needs. Projection is therefore not only a safeguard but a component of the certificate.

Remark 5 (Actuator compatibility). *The relevant hardware constraint is on the whole compensation, not on its error. If $\tau_{\max}$ is the available torque and the robust action must retain authority $\nu_{\max}$, then $\bar{\Delta}$ must satisfy $\|\tilde{W}\Sigma\| \leq \bar{\Delta}$ with $\bar{\Delta} + \nu_{\max} + \|\tau_{\text{pbc}}\| \leq \tau_{\max}$. A network that is accurate but saturates the actuator is useless, because the robust action loses the margin it needs to dominate $\tilde{\Delta}$. This condition will be checked explicitly on the experimental platform.*

5.5 Robust Action and Gain Condition

Provenance. The algorithm is Levant’s super-twisting [8, 9], the strict Lyapunov function is that of [13], the matrix-gain treatment that of [14], and the variable-gain fallback that of [4]. *What is modified.* All of these assume the bound $\|\frac{d}{dt}\tilde{\Delta}\| \leq L_\theta$. Here L_θ is produced by the learning architecture instead, through the three terms of (12); and the storage function (19) composes the form of [13] with the passive storage of the nominal loop and the parameter energy of the adaptation, which is what allows the coupling $M^{-1}(q)$ to be dominated using (2).

The Super-Twisting algorithm is applied to the variable s :

$$\tau_{\text{sta}} = -K_1 [s]^{1/2} + v, \quad \dot{v} = -K_2 [s]^0, \quad [x]^p := |x|^p \operatorname{sign}(x) \text{ (componentwise)}. \quad (18)$$

Since s has relative degree one, τ_{sta} is continuous, which attenuates chattering relative to a first-order sliding mode.

Assumption 1. *There exist constants $\delta_\theta, L_\theta > 0$ such that, on the compact operating set, $\|\tilde{\Delta}\| \leq \delta_\theta$ and $\|\frac{d}{dt}\tilde{\Delta}\| \leq L_\theta$.*

The second condition is the *essential* restriction of the STA and is usually omitted in the neural control literature. Two routes will be explored to satisfy it in a non-circular manner: (i) certifying L_θ from the Lipschitz constant of the network, controlled by spectral normalization and projection, together with the growth bounds (5) and the bounds on $\dot{q}, \ddot{q}_r$ over the compact operating set — the same inheritance argument by which the approximation error of a Lipschitz plant is itself Lipschitz; and (ii) employing a variable-gain version of the STA, in which K_1, K_2 depend on an online known bound of the disturbance.

The stability analysis will combine the passive storage function of the nominal loop, the parameter energy of (17) and the strict Lyapunov function of [13] for the STA:

$$V(s, v, \tilde{W}) = \frac{1}{2} s^\top M(q) s + \sum_{i=1}^n \zeta_i^\top P_i \zeta_i + \frac{1}{2} \text{tr}(\tilde{W}^\top K_w^{-1} \tilde{W}), \quad \zeta_i = [s_i]^{1/2}, v_i]^\top, \quad (19)$$

where the first term provides the passive structure, the second finite-time convergence and the third the exact cancellation of the adaptation. The technical contribution consists in establishing the conditions on K_1, K_2, K_D, K_w and λ_m, λ_M under which $\dot{V}$ is negative definite despite the coupling $M^{-1}(q)$.

5.6 Honest Quantification of the Benefit of Learning

Provenance. The residual-set picture — learning shrinks the ball, an integral action removes it — is the conclusion of the neural PD/PID line [10, 29, 30]. *What is modified.* There the integrator removes a *constant* residual asymptotically; here the STA state v removes a *time-varying* residual in finite time. The consequence is that the benefit of learning must be restated: it is no longer accuracy, but gain magnitude, effort, noise transmission and sampled accuracy. This is contribution (N3).

Remark 6 (Necessary conceptual precision). *Under the ideal STA with a matched, Lipschitz disturbance and correctly sized gains, one has $s \equiv 0$ in finite time and consequently $e \rightarrow 0$ regardless of the magnitude of δ_θ . Therefore, **learning cannot improve the asymptotic tracking accuracy in the ideal case**, and claiming otherwise would be incorrect. The benefit of learning must be formulated in the metrics where it is in fact demonstrable.*

The classical gain conditions are of the form $k_2 > c_1 L_\theta$ and $k_1 > c_2 \sqrt{L_\theta}$. From these, four verifiable benefits follow:

- (B1) **Gain reduction.** If learning reduces the Lipschitz bound by a factor $\rho \in (0, 1)$, then k_2 is reduced by ρ and k_1 by $\sqrt{\rho}$.
- (B2) **Control effort and smoothness.** Smaller gains imply a smaller $\int \|\tau\|^2$ and a smaller total variation of the torque, that is, less chattering and less actuator wear.
- (B3) **Noise sensitivity.** The amplification of measurement noise in the term $[s]^{1/2}$ scales with k_1 ; reducing it improves performance when velocity is estimated by encoder differentiation.
- (B4) **Real sliding accuracy under sampling.** In a discrete implementation with period h , the attainable accuracy satisfies $|s| = O(k^2 h^2)$. *This is the mechanism through which learning does improve the practical tracking error*, and it constitutes the most defensible result of the proposal.

Remark 7 (The robust action as a nonlinear integrator). *The state v of (18) obeys $\dot{v} = -K_2[s]^0$ and converges in finite time to $\tilde{\Delta}$: it is an integrator whose steady value is exactly the residual the module failed to reproduce. This is the same division of labour found in neural PID control of manipulators, where the network carries $g + F$ and the integrator carries only the approximation error, so that the integral gain can be two orders of magnitude below the proportional gain. The proposal transfers that observation to the sliding mode setting: the module carries $\Delta_{\text{el}} + \Delta_{\text{dis}}$ and the STA carries only $\tilde{\Delta}$, which is why K_1, K_2 can be small. The distinction with the neural PID case is that here the “integrator” is discontinuous in s and therefore rejects a time-varying $\tilde{\Delta}$, not merely a constant one.*

5.7 Velocity Estimation without Differentiation

Provenance. The first-order, zero-relative-degree filter realized without differentiation is the observer used in neural PID control of manipulators [29]; the exact alternative is Levant’s differentiator [9].

What is modified. The filter is chosen to coincide with the training operator $\mathcal{F}$ of (14), so the same dynamic map is used for identification and for feedback, and its noise transmission into $[s]^{1/2}$ is measured rather than assumed.

The gain reduction (B3) is only useful if the velocity signal is itself clean. Encoders give excellent positions and poor velocities, so $\dot{q}$ will be replaced, where needed, by the output of a first-order zero-relative-degree filter $v_i(s) = \frac{b_i s}{s + a_i} q_i(s)$, realized without differentiation as

$$\dot{z} = -A(z + Bq), \quad v = z + Bq, \quad (20)$$

with A, B positive diagonal. This is the same filtering operator used in (14), which keeps training and implementation consistent. The alternative — the robust exact differentiator of [9] — will be compared against (20) in terms of the order of accuracy attained and the noise transmitted to $[s]^{1/2}$.

5.8 Design Recipe

The analysis produces an explicit order of decisions, which will be validated experimentally:

1. Bound the inertia without a model: $\lambda_M \leq \beta$ with $\beta \geq n \max_{i,j} |m_{ij}|$; identify k_c, k_g from the geometry.
2. Tune K_D and Λ as a stable passivity-based loop with the module disabled ($\hat{W} = 0$) and the robust action off; record the baseline radius ρ_1 .
3. Train the module offline on the persistent-excitation data set; measure δ_θ and estimate L_θ on the validation partition.
4. Set $\bar{\Delta}$ from the actuator budget and enable projection.
5. Read $k_1^{\min}, k_2^{\min}$ from the certified $(\delta_\theta, L_\theta)$; select gains with a small margin above the minimum rather than a worst-case margin.
6. Enable the online law (17) with a small K_w and increase it until the residual falls without oscillation.

5.9 Experimental Validation

The strategy will be implemented on a manipulator platform available in the laboratory, with real-time acquisition and a fixed sampling period. Four configurations will be evaluated in an ablation study: (C1) nominal PBC; (C2) PBC + learned compensation; (C3) PBC + STA with worst-case gains; (C4) complete architecture. Metrics: RMSE and ITAE of the joint error, $\int \|\tau\|^2$, total variation of τ , chattering index, time to reach the surface, noise transmitted to the control, and the magnitude of the admissible gains. Disturbance scenarios not seen during training (additional payload, contact) will be included in order to assess generalization and to verify that the degradation is bounded — in particular, that a saturated or badly initialized module degrades performance without destabilizing the loop.

6 Expected Contributions

The first three items are the contributions proper, in the sense of Section 2; the last three are the supporting results without which the first three cannot be stated or tested.

- C1. **Derivative certificate (N1).** An *a priori* bound on $\|\frac{d}{dt}\tilde{\Delta}\|$ along closed-loop trajectories, obtained by bounding the three terms of (12) — plant growth via (5), network Lipschitz constant via spectral normalization and projection, and adaptation rate via the learning law itself — so that the super-twisting gain condition is closed non-circularly rather than assumed.
- C2. **Structure-preserving adaptation (N2).** An admissible weight set $\mathcal{W}$ and a projected Lyapunov law whose trace term cancels exactly, such that the Euler–Lagrange structural properties, and hence the output strict passivity of the loop, hold for *every* admissible weight and therefore along every trajectory while learning is active.
- C3. **Correct and quantified statement of the benefit (N3).** A proof that learning cannot improve ideal asymptotic accuracy, together with the explicit map $(\delta_\theta, L_\theta) \mapsto (k_1^{\min}, k_2^{\min})$ and the characterization of the improvement in real sliding accuracy $|s| = O(k^2 h^2)$ under sampling.
- C4. **Supporting: decomposition of the residual.** The splitting (11), which delimits formally the scope of learning against that of the robust action and fixes, before any experiment, what each mechanism is responsible for.
- C5. **Supporting: design recipe.** An explicit order of decisions in which every inequality uses only λ_M, k_c, k_g and the actuator limits, so that the loop can be certified without an identified dynamic model.
- C6. **Supporting: experimental validation.** Ablation C1–C4 on a physical platform, reporting *certified* and *measured* L_θ side by side, so that the conservatism of C1 is itself a reported quantity.

7 Work Plan

First four-month term — Fundamentals, modeling and nominal control

Literature review on passivity-based control, manipulator dynamics, neural compensation of PD/PID loops, structured learning of mechanical systems and higher-order sliding modes. Design of the nominal controller and its stability analysis, including the baseline radius (10). Precise formulation of the residual dynamics (9) and of its decomposition (11). Commissioning of the experimental platform and identification of the nominal model.

Deliverables: identified nominal model; nominal controller running on hardware; state-of-the-art document.

Second four-month term — Learning, robust action and analysis

Construction of module (13) and of the acceleration-free training scheme; design of excitation trajectories and data collection. Derivation of the online law (17) and proof of passivity preservation under projection. Design of the multivariable STA and derivation of the gain conditions from (19). Establishment of the relation $(\delta_\theta, L_\theta) \mapsto (k_1^{\min}, k_2^{\min})$. Validation in high-fidelity simulation.

Deliverables: stability and finite-time convergence theorems; simulation results; draft of a conference paper.

Third four-month term — Experimental validation and writing

Full experimental implementation and ablation study C1–C4. Comparison of the results with the theoretical predictions, with emphasis on the relation between learning error, required gains and real sliding accuracy. Integration of results, writing of the thesis and preparation of academic outputs.

Deliverables: experimental results; thesis document; journal paper submitted.

Risks and contingency plan

- *Delay in the experimental platform.* Alternative: validation in a high-fidelity simulator with identified parameters and realistic sensor noise, reserving the hardware for a reduced two-degree-of-freedom demonstration.
- *Overly conservative Lipschitz certificate.* Alternative: variable-gain STA with online estimation of the bound, which relaxes the need for a tight *a priori* certificate.
- *Noise in velocity estimation.* Alternative: the filter (20), the robust exact differentiator of [9], or an adaptive filter, evaluating the effect of each on the order of accuracy attained.
- *Drift of the online adaptation.* Alternative: the robust modifications of Section 4.4, with the dead zone sized from the measured noise floor.

8 Expected Academic Outputs

One peer-reviewed international conference paper (IEEE CDC, IFAC World Congress or IEEE CASE) on the formulation and the stability analysis, and one journal paper (*IEEE Transactions on Control Systems Technology*, *IEEE/ASME Transactions on Mechatronics* or *Neurocomputing*) with the complete experimental validation.

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